COHOMOLOGICAL DESCENT FOR ARBITRARY DERIVED OBJECTS ALONG A FIXED HYPERCOVER

THE STACKS PROJECT AUTHORS

Editorial note. Despite the historical label containingbounded-abelian, the exact statement is forE inD(C) with no boundedness hypothesis. That full scope and author proof are retained.

Editorial note. The ordinary hypercoverCech-colimit theorem is not supplied as a shortcut to the unbounded unit isomorphism.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Sites with equalizers and fibre products, total sites of simplicial objects, K-injective resolutions, derived adjunction and product total complexes. The local sheaf-descent,Cech-complex and augmentation proofs and the hypercover acyclicity lemma are retained.

Source reference. Original section 09WE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12492–12492. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0D6Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16539–16539. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D79: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16549–16549. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09WG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12494–12494. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09WH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12495–12495. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0132: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1420–1420. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

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concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0133: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1421–1421. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03F3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4478–4478. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08FI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10711–10711. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D87: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16582–16582. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 07A0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9290–9290. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D7H: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16557–16557. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D86: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16581–16581. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 04TP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6262–6262. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D7I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16558–16558. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 03CI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4385–4385. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 0165: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1532–1532. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0187: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1606–1606. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04TQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6263–6263. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04TS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6265–6265. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01GB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1900–1900. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01GE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1903–1903. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 015E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1505–1505. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D78: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16548–16548. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 08J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10840–10840. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0A94: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12937–12937. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 08JA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10843–10843. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 0EYY: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18708–18708. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0EYX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 18707–18707. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 016E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1541–1541. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 018J: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1618–1618. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 018I: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1617–1617. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01GC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1901–1901. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 01GD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1902–1902. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 01G0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1889–1889. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 09WK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12498–12498. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original section 09WB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12489–12489. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D85: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16580–16580. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 09WD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12491–12491. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D70: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16540–16540. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 00XR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1229–1229. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 04BG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5651–5651. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0DBZ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16714–16714. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0797: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9262–9262. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two

concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original lemma 0D74: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16544–16544. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Source reference. Original definition 09VU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 12472–12472. Stable Stacks tag for an expressly allowed foundational or notational input; the counted construction remains the two concrete core obligations. The retained local core is not replaced by this pointer. See standard_prerequisites.

Lemma 0.1. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. For $E \in D(\mathcal{C})$ the map*

$$E \longrightarrow Ra_*a^{-1}E$$

is an isomorphism.

Remark 0.2 (Variant for over an object). Let $\mathcal{C}$ be a site. Let $X \in \mathrm{Ob}(\mathcal{C})$. The category $\mathrm{SR}(\mathcal{C}, X)$ of *semi-representable objects over X* is defined by the formula $\mathrm{SR}(\mathcal{C}, X) = \mathrm{SR}(\mathcal{C}/X)$. See Hypercoverings, Definition 01G0. Thus we may apply the above discussion to the site $\mathcal{C}/X$. Briefly, the constructions above give

(1) a site $\mathcal{C}/K$ for K in $\mathrm{SR}(\mathcal{C}, X)$,
(2) a decomposition $Sh(\mathcal{C}/K) = \prod Sh(\mathcal{C}/U_i)$ if $K = \{U_i/X\}$,
(3) a localization functor $j : \mathcal{C}/K \to \mathcal{C}/X$,
(4) a morphism $f : Sh(\mathcal{C}/K) \to Sh(\mathcal{C}/L)$ for $f : K \to L$ in $\mathrm{SR}(\mathcal{C}, X)$.

All results of this section hold in this situation by replacing $\mathcal{C}$ everywhere by $\mathcal{C}/X$.

Remark 0.3 (Ringed variant). Let $\mathcal{C}$ be a site. Let $\mathcal{O}_\mathcal{C}$ be a sheaf of rings on $\mathcal{C}$. In this case, for any semi-representable object K of $\mathcal{C}$ the site $\mathcal{C}/K$ is a ringed site with sheaf of rings $\mathcal{O}_K = j^{-1}\mathcal{O}_\mathcal{C}$. The constructions above give

(1) a ringed site $(\mathcal{C}/K, \mathcal{O}_K)$ for K in $\mathrm{SR}(\mathcal{C})$,
(2) a decomposition $Mod(\mathcal{O}_K) = \prod Mod(\mathcal{O}_{U_i})$ if $K = \{U_i\}$,
(3) a localization morphism $j : (Sh(\mathcal{C}/K), \mathcal{O}_K) \to (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C})$ of ringed topoi,
(4) a morphism $f : (Sh(\mathcal{C}/K), \mathcal{O}_K) \to (Sh(\mathcal{C}/L), \mathcal{O}_L)$ of ringed topoi for $f : K \to L$ in $\mathrm{SR}(\mathcal{C})$.

Many of the results above hold in this setting. For example, the functor j^* has an exact left adjoint

$$j_! : Mod(\mathcal{O}_K) \to Mod(\mathcal{O}_\mathcal{C}),$$

which in terms of the product decomposition given in (2) sends $(\mathcal{F}_i)_{i\in I}$ to $\bigoplus j_{i,!}\mathcal{F}_i$. Similarly, given $f : K \to L$ as above, the functor f^* has an exact left adjoint $f_! : Mod(\mathcal{O}_K) \to Mod(\mathcal{O}_L)$. Thus the functors j^* and f^* are exact, i.e., j and f are flat morphisms of ringed topoi (also follows from the equalities $\mathcal{O}_K = j^{-1}\mathcal{O}_\mathcal{C}$ and $\mathcal{O}_K = f^{-1}\mathcal{O}_L$).

Remark 0.4 (Ringed variant over an object). Let $\mathcal{C}$ be a site. Let $\mathcal{O}_\mathcal{C}$ be a sheaf of rings on $\mathcal{C}$. Let $X \in \mathrm{Ob}(\mathcal{C})$ and denote $\mathcal{O}_X = \mathcal{O}_\mathcal{C}|_{\mathcal{C}/U}$. Then we can combine the constructions given in Remarks 0.2 and 0.3 to get

(1) a ringed site $(\mathcal{C}/K, \mathcal{O}_K)$ for K in $\mathrm{SR}(\mathcal{C}, X)$,
(2) a decomposition $Mod(\mathcal{O}_K) = \prod Mod(\mathcal{O}_{U_i})$ if $K = \{U_i\}$,
(3) a localization morphism $j : (Sh(\mathcal{C}/K), \mathcal{O}_K) \to (Sh(\mathcal{C}/X), \mathcal{O}_X)$ of ringed topoi,

(4) a morphism $f : (Sh(\mathcal{C}/K), \mathcal{O}_K) \to (Sh(\mathcal{C}/L), \mathcal{O}_L)$ of ringed topoi for $f : K \to L$ in $\mathrm{SR}(\mathcal{C}, X)$.

Of course all of the results mentioned in Remark 0.3 hold in this setting as well.

1. The site associate to a simplicial semi-representable object

Let $\mathcal{C}$ be a site. Let K be a simplicial object of $\mathrm{SR}(\mathcal{C})$. As usual, set $K_n = K([n])$ and denote $K(\varphi) : K_n \to K_m$ the morphism associated to $\varphi : [m] \to [n]$. By the construction in Section 09WK we obtain a simplicial object $n \mapsto \mathcal{C}/K_n$ in the category whose objects are sites and whose morphisms are cocontinuous functors. In other words, we get a gadget as in Case B of Section 09WB. The functors satisfy property P by Lemma 0D85. Hence we may apply Lemma 09WD to obtain a site $(\mathcal{C}/K)_{total}$.

We can describe the site $(\mathcal{C}/K)_{total}$ explicitly as follows. Say $K_n = \{U_{n,i}\}_{i \in I_n}$. For $\varphi : [m] \to [n]$ the morphism $K(\varphi) : K_n \to K_m$ is given by a map $\alpha(\varphi) : I_n \to I_m$ and morphisms $f_{\varphi,i} : U_{n,i} \to U_{m,\alpha(\varphi)(i)}$ for $i \in I_n$. Then we have

(1) an object of $(\mathcal{C}/K)_{total}$ corresponds to an object $(U/U_{n,i})$ of $\mathcal{C}/U_{n,i}$ for some n and some $i \in I_n$,
(2) a morphism between $U/U_{n,i}$ and $V/U_{m,j}$ is a pair (φ, f) where $\varphi : [m] \to [n]$, $j = \alpha(\varphi)(i)$, and $f : U \to V$ is a morphism of $\mathcal{C}$ such that

$$\begin{array}{ccc} U & \xrightarrow{f} & V \\ \downarrow & & \downarrow \\ U_{n,i} & \xrightarrow{f_{\varphi,i}} & U_{m,j} \end{array}$$

is commutative, and
(3) coverings of the object $U/U_{n,i}$ are constructed by starting with a covering $\{f_j : U_j \to U\}$ in $\mathcal{C}$ and letting $\{(\mathrm{id}, f_j) : U_j/U_{n,i} \to U/U_{n,i}\}$ be a covering in $(\mathcal{C}/K)_{total}$.

All of our general theory developed for simplicial sites applies to $(\mathcal{C}/K)_{total}$. Observe that the obvious forgetful functor

$$j_{total} : (\mathcal{C}/K)_{total} \longrightarrow \mathcal{C}$$

is continuous and cocontinuous. It turns out that the associated morphism of topoi comes from an (obvious) augmentation.

Lemma 1.1. *Let $\mathcal{C}$ be a site. Let K be a simplicial object of $SR(\mathcal{C})$. The localization functor $j_0 : \mathcal{C}/K_0 \to \mathcal{C}$ defines an augmentation $a_0 : Sh(\mathcal{C}/K_0) \to Sh(\mathcal{C})$, as in case (B) of Remark 0D6Z. The corresponding morphisms of topoi*

$$a_n : Sh(\mathcal{C}/K_n) \longrightarrow Sh(\mathcal{C}), \quad a : Sh((\mathcal{C}/K)_{total}) \longrightarrow Sh(\mathcal{C})$$

of Lemma 0D70 are equal to the morphisms of topoi associated to the continuous and cocontinuous localization functors $j_n : \mathcal{C}/K_n \to \mathcal{C}$ and $j_{total} : (\mathcal{C}/K)_{total} \to \mathcal{C}$.

Proof. This is immediate from working through the definitions. See in particular the footnote in the proof of Lemma 0D70 for the relationship between a and j_{total}. □

Lemma 1.2. *With assumption and notation as in Lemma 1.1 we have the following properties:*

(1) *there is a functor $a_!^{Sh} : Sh((\mathcal{C}/K)_{total}) \to Sh(\mathcal{C})$ left adjoint to $a^{-1} : Sh(\mathcal{C}) \to Sh((\mathcal{C}/K)_{total})$,*
(2) *there is a functor $a_! : Ab((\mathcal{C}/K)_{total}) \to Ab(\mathcal{C})$ left adjoint to $a^{-1} : Ab(\mathcal{C}) \to Ab((\mathcal{C}/K)_{total})$,*

(3) *the functor* a^{-1} *associates to* $\mathcal{F}$ *in* $Sh(\mathcal{C})$ *the sheaf on* $(\mathcal{C}/K)_{total}$ *which in degree* n *is equal to* $a_n^{-1}\mathcal{F}$,
(4) *the functor* a_* *associates to* $\mathcal{G}$ *in* $Ab((\mathcal{C}/K)_{total})$ *the equalizer of the two maps* $j_{0,*}\mathcal{G}_0 \to j_{1,*}\mathcal{G}_1$,

Proof. Parts (3) and (4) hold for any augmentation of a simplicial site, see Lemma 0D70. Parts (1) and (2) follow as j_{total} is continuous and cocontinuous. The functor $a_!^{Sh}$ is constructed in Sites, Lemma 00XR and the functor $a_!$ is constructed in Modules on Sites, Lemma 04BG. □

Lemma 1.3. *Let* $\mathcal{C}$ *be a site. Let* K *be a simplicial object of* $SR(\mathcal{C})$. *Let* $U/U_{n,i}$ *be an object of* $\mathcal{C}/K_n$. *Let* $\mathcal{F} \in Ab((\mathcal{C}/K)_{total})$. *Then*

$$H^p(U, \mathcal{F}) = H^p(U, \mathcal{F}_{n,i})$$

where

(1) *on the left hand side* U *is viewed as an object of* $\mathcal{C}_{total}$, *and*
(2) *on the right hand side* $\mathcal{F}_{n,i}$ *is the* i*th component of the sheaf* $\mathcal{F}_n$ *on* $\mathcal{C}/K_n$ *in the decomposition* $Sh(\mathcal{C}/K_n) = \prod Sh(\mathcal{C}/U_{n,i})$ *of Section 09WK.*

Proof. This follows immediately from Lemma 0DBZ and the product decompositions of Section 09WK. □

Remark 1.4 (Variant for over an object). Let $\mathcal{C}$ be a site. Let $X \in \mathrm{Ob}(\mathcal{C})$. Recall that we have a category $\mathrm{SR}(\mathcal{C}, X) = \mathrm{SR}(\mathcal{C}/X)$ of semi-representable objects over X, see Remark 0.2. We may apply the above discussion to the site $\mathcal{C}/X$. Briefly, the constructions above give

(1) a site $(\mathcal{C}/K)_{total}$ for a simplicial K object of $\mathrm{SR}(\mathcal{C}, X)$,
(2) a localization functor $j_{total} : (\mathcal{C}/K)_{total} \to \mathcal{C}/X$,
(3) localization functors $j_n : \mathcal{C}/K_n \to \mathcal{C}/X$,
(4) a morphism of topoi $a : Sh((\mathcal{C}/K)_{total}) \to Sh(\mathcal{C}/X)$,
(5) morphisms of topoi $a_n : Sh(\mathcal{C}/K_n) \to Sh(\mathcal{C}/X)$,
(6) a functor $a_!^{Sh} : Sh((\mathcal{C}/K)_{total}) \to Sh(\mathcal{C}/X)$ left adjoint to a^{-1}, and
(7) a functor $a_! : Ab((\mathcal{C}/K)_{total}) \to Ab(\mathcal{C}/X)$ left adjoint to a^{-1}.

All of the results of this section hold in this setting. To prove this one replaces the site $\mathcal{C}$ everywhere by $\mathcal{C}/X$.

Remark 1.5 (Ringed variant). Let $\mathcal{C}$ be a site. Let $\mathcal{O}_\mathcal{C}$ be a sheaf of rings. Given a simplicial semi-representable object K of $\mathcal{C}$ we set $\mathcal{O} = a^{-1}\mathcal{O}_\mathcal{C}$, where a is as in Lemmas 1.1 and 1.2. The constructions above, keeping track of the sheaves of rings as in Remark 0.3, give

(1) a ringed site $((\mathcal{C}/K)_{total}, \mathcal{O})$ for a simplicial K object of $\mathrm{SR}(\mathcal{C})$,
(2) a morphism of ringed topoi $a : (Sh((\mathcal{C}/K)_{total}), \mathcal{O}) \to (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C})$,
(3) morphisms of ringed topoi $a_n : (Sh(\mathcal{C}/K_n), \mathcal{O}_n) \to (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C})$,
(4) a functor $a_! : Mod(\mathcal{O}) \to Mod(\mathcal{O}_\mathcal{C})$ left adjoint to a^*.

The functor $a_!$ exists (but in general is not exact) because $a^{-1}\mathcal{O}_\mathcal{C} = \mathcal{O}$ and we can replace the use of Modules on Sites, Lemma 04BG in the proof of Lemma 1.2 by Modules on Sites, Lemma 0797. As discussed in Remark 0.3 there are exact functors $a_{n!} : Mod(\mathcal{O}_n) \to Mod(\mathcal{O}_\mathcal{C})$ left adjoint to a_n^*. Consequently, the morphisms a and a_n are flat. Remark 0.3 implies the morphism of ringed topoi $f_\varphi : (Sh(\mathcal{C}/K_n), \mathcal{O}_n) \to (Sh(\mathcal{C}/K_m), \mathcal{O}_m)$ for $\varphi : [m] \to [n]$ is flat and there exists an exact functor $f_{\varphi!} : Mod(\mathcal{O}_n) \to Mod(\mathcal{O}_m)$ left adjoint to f_φ^*. This in turn implies that for the flat morphism of ringed topoi $g_n : (Sh(\mathcal{C}/K_n), \mathcal{O}_n) \to (Sh((\mathcal{C}/K)_{total}), \mathcal{O})$ the functor $g_{n!} : Mod(\mathcal{O}_n) \to Mod(\mathcal{O})$ left adjoint to g_n^* is exact, see Lemma 0D74.

Remark 1.6 (Ringed variant over an object). Let $\mathcal{C}$ be a site. Let $\mathcal{O}_\mathcal{C}$ be a sheaf of rings. Let $X \in \mathrm{Ob}(\mathcal{C})$ and denote $\mathcal{O}_X = \mathcal{O}_\mathcal{C}|_{\mathcal{C}/X}$. Then we can combine the constructions given in Remarks 1.4 and 1.5 to get

(1) a ringed site $((\mathcal{C}/K)_{total}, \mathcal{O})$ for a simplicial K object of $\mathrm{SR}(\mathcal{C}, X)$,
(2) a morphism of ringed topoi $a : (Sh((\mathcal{C}/K)_{total}), \mathcal{O}) \to (Sh(\mathcal{C}/X), \mathcal{O}_X)$,
(3) morphisms of ringed topoi $a_n : (Sh(\mathcal{C}/K_n), \mathcal{O}_n) \to (Sh(\mathcal{C}/X), \mathcal{O}_X)$,
(4) a functor $a_! : Mod(\mathcal{O}) \to Mod(\mathcal{O}_X)$ left adjoint to a^*.

Of course, all the results mentioned in Remark 1.5 hold in this setting as well.

2. Cohomological descent for hypercoverings

Let $\mathcal{C}$ be a site. In this section we assume $\mathcal{C}$ has equalizers and fibre products. We let K be a hypercovering as defined in Hypercoverings, Definition 09VU. We will study the augmentation

$$a : Sh((\mathcal{C}/K)_{total}) \longrightarrow Sh(\mathcal{C})$$

of Section 1.

Lemma 2.1. *In Situation 09WE let a_0 be an augmentation towards a site $\mathcal{D}$ as in Remark 0D6Z. For any K in $D^+(\mathcal{C}_{total})$ there is a spectral sequence $(E_r, d_r)_{r\geq 0}$ with*

$$E_1^{p,q} = R^q a_{p,*} K_p, \quad d_1^{p,q} : E_1^{p,q} \to E_1^{p+1,q}$$

converging to $R^{p+q}a_*K$. This spectral sequence is functorial in K.

Proof. Let $\mathcal{I}^\bullet$ be a bounded below complex of injectives representing K. Consider the double complex with terms

$$A^{p,q} = a_{p,*}\mathcal{I}_p^q$$

where the horizontal arrows come from Lemma 0D79 and the vertical arrows from the differentials of the complex $\mathcal{I}^\bullet$. The rows of the double complex are exact in positive degrees and evaluate to $a_*\mathcal{I}^q$ in degree 0. On the other hand, since restriction to $\mathcal{C}_p$ is exact (Lemma 09WG) the complex $\mathcal{I}_p^\bullet$ represents K_p in $D(\mathcal{C}_p)$. The sheaves $\mathcal{I}_p^q$ are injective abelian sheaves on $\mathcal{C}_p$ (Lemma 09WH). Hence the cohomology of the columns computes $R^q a_{p,*} K_p$. We conclude by applying Homology, Lemmas 0132 and 0133. □

Lemma 2.2. *Let $\mathcal{C}$ be a site. For K in $SR(\mathcal{C})$ the functor j^{-1} sends injective abelian sheaves to injective abelian sheaves. Similarly, the functor j^{-1} sends K-injective complexes of abelian sheaves to K-injective complexes of abelian sheaves.*

Proof. The first statement is the natural generalization of Cohomology on Sites, Lemma 03F3 to semi-representable objects. In fact, it follows from this lemma by the product decomposition of $Sh(\mathcal{C}/K)$ and the description of the functor j^{-1} given above. The second statement is the natural generalization of Cohomology on Sites, Lemma 08FI and follows from it by the product decomposition of the topos.

Alternative: since j induces a localization of topoi by Lemma 0D87 part (1) it also follows immediately from Cohomology on Sites, Lemmas 03F3 and 08FI by enlarging the site; compare with the proof of Cohomology on Sites, Lemma 07A0 in the case of injective sheaves. □

Lemma 2.3. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. Then*

(1) *$a^{-1} : Sh(\mathcal{C}) \to Sh((\mathcal{C}/K)_{total})$ is fully faithful with essential image the cartesian sheaves of sets,*

(2) $a^{-1} : Ab(\mathcal{C}) \to Ab((\mathcal{C}/K)_{total})$ *is fully faithful with essential image the cartesian sheaves of abelian groups.*

In both cases a_* provides the quasi-inverse functor.

Proof. The case of abelian sheaves follows immediately from the case of sheaves of sets as the functor a^{-1} commutes with products. In the rest of the proof we work with sheaves of sets. Observe that $a^{-1}\mathcal{F}$ is cartesian for $\mathcal{F}$ in $Sh(\mathcal{C})$ by Lemma 0D7H. It suffices to show that the adjunction map $\mathcal{F} \to a_*a^{-1}\mathcal{F}$ is an isomorphism $\mathcal{F}$ in $Sh(\mathcal{C})$ and that for a cartesian sheaf $\mathcal{G}$ on $(\mathcal{C}/K)_{total}$ the adjunction map $a^{-1}a_*\mathcal{G} \to \mathcal{G}$ is an isomorphism.

Let $\mathcal{F}$ be a sheaf on $\mathcal{C}$. Recall that $a_*a^{-1}\mathcal{F}$ is the equalizer of the two maps $a_{0,*}a_0^{-1}\mathcal{F} \to a_{1,*}a_1^{-1}\mathcal{F}$, see Lemma 1.2. By Lemma 0D86

$$a_{0,*}a_0^{-1}\mathcal{F} = \mathcal{H}om(F(K_0)^\#, \mathcal{F}) \quad \text{and} \quad a_{1,*}a_1^{-1}\mathcal{F} = \mathcal{H}om(F(K_1)^\#, \mathcal{F})$$

On the other hand, we know that

$$F(K_1)^\# \rightrightarrows F(K_0)^\# \longrightarrow \text{final object } * \text{ of } Sh(\mathcal{C})$$

is a coequalizer diagram in sheaves of sets by definition of a hypercovering. Thus it suffices to prove that $\mathcal{H}om(-, \mathcal{F})$ transforms coequalizers into equalizers which is immediate from the construction in Sites, Section 04TP.

Let $\mathcal{G}$ be a cartesian sheaf on $(\mathcal{C}/K)_{total}$. We will show that $\mathcal{G} = a^{-1}\mathcal{F}$ for some sheaf $\mathcal{F}$ on $\mathcal{C}$. This will finish the proof because then $a^{-1}a_*\mathcal{G} = a^{-1}a_*a^{-1}\mathcal{F} = a^{-1}\mathcal{F} = \mathcal{G}$ by the result of the previous paragraph. Set $\mathcal{K}_n = F(K_n)^\#$ for $n \geq 0$. Then we have maps of sheaves

$$\mathcal{K}_2 \rightrightarrows \mathcal{K}_1 \rightrightarrows \mathcal{K}_0$$

coming from the fact that K is a simplicial semi-representable object. The fact that K is a hypercovering means that

$$\mathcal{K}_1 \to \mathcal{K}_0 \times \mathcal{K}_0 \quad \text{and} \quad \mathcal{K}_2 \to \Big(\text{cosk}_1(\ \mathcal{K}_1 \rightrightarrows \mathcal{K}_0\ )\Big)_2$$

are surjective maps of sheaves. Using the description of cartesian sheaves on $(\mathcal{C}/K)_{total}$ given in Lemma 0D7I and using the description of $Sh(\mathcal{C}/K_n)$ in Lemma 0D87 we find that our problem can be entirely formulated[1] in terms of

(1) the topos $Sh(\mathcal{C})$, and
(2) the simplicial object $\mathcal{K}$ in $Sh(\mathcal{C})$ whose terms are $\mathcal{K}_n$.

Thus, after replacing $\mathcal{C}$ by a different site $\mathcal{C}'$ as in Sites, Lemma 03CI, we may assume $\mathcal{C}$ has all finite limits, the topology on $\mathcal{C}$ is subcanonical, a family $\{V_j \to V\}$ of morphisms of $\mathcal{C}$ is a covering if and only if $\coprod h_{V_j} \to V$ is surjective, and there exists a simplicial object U of $\mathcal{C}$ such that $\mathcal{K}_n = h_{U_n}$ as simplicial sheaves. Working backwards through the equivalences we may assume $K_n = \{U_n\}$ for all n.

Let X be the final object of $\mathcal{C}$. Then $\{U_0 \to X\}$ is a covering, $\{U_1 \to U_0 \times U_0\}$ is a covering, and $\{U_2 \to (\text{cosk}_1\text{sk}_1U)_2\}$ is a covering. Let us use $d_i^n : U_n \to U_{n-1}$ and $s_j^n : U_n \to U_{n+1}$ the morphisms corresponding to δ_i^n and σ_j^n as in Simplicial, Definition 0165. By abuse of notation, given a morphism $c : V \to W$ of $\mathcal{C}$ we denote the morphism of topoi $c : Sh(\mathcal{C}/V) \to Sh(\mathcal{C}/W)$ by the same letter. Now $\mathcal{G}$ is given by a sheaf $\mathcal{G}_0$ on $\mathcal{C}/U_0$ and an isomorphism $\alpha : (d_1^1)^{-1}\mathcal{G}_0 \to (d_0^1)^{-1}\mathcal{G}_0$ satisfying the

[1]Even though it does not matter what the precise formulation is, we spell it out: the problem is to show that given an object $\mathcal{G}_0/\mathcal{K}_0$ of $Sh(\mathcal{C})/\mathcal{K}_0$ and an isomorphism

$$\alpha : \mathcal{G}_0 \times_{\mathcal{K}_0, \mathcal{K}(\delta_1^1)} \mathcal{K}_1 \to \mathcal{G}_0 \times_{\mathcal{K}_0, \mathcal{K}(\delta_0^1)} \mathcal{K}_1$$

over $\mathcal{K}_1$ satisfying a cocycle condition in $Sh(\mathcal{C})/\mathcal{K}_2$, there exists $\mathcal{F}$ in $Sh(\mathcal{C})$ and an isomorphism $\mathcal{F} \times \mathcal{K}_0 \to \mathcal{G}_0$ over $\mathcal{K}_0$ compatible with α.

cocycle condition on $\mathcal{C}/U_2$ formulated in Lemma 0D7I. Since $\{U_2 \to (\mathrm{cosk}_1\mathrm{sk}_1 U)_2\}$ is a covering, the corresponding pullback functor on sheaves is faithful (small detail omitted). Hence we may replace U by $\mathrm{cosk}_1\mathrm{sk}_1 U$, because this replaces U_2 by $(\mathrm{cosk}_1\mathrm{sk}_1 U)_2$ and leaves U_1 and U_0 unchanged. Then

$$(d_0^2, d_1^2, d_2^2) : U_2 \to U_1 \times U_1 \times U_1$$

is a monomorphism whose its image on T-valued points is described in Simplicial, Lemma 0187. In particular, there is a morphism c fitting into a commutative diagram

$$\begin{array}{ccc} U_1 \times_{(d_1^1, d_0^1), U_0 \times U_0, (d_1^1, d_0^1)} U_1 & \xrightarrow[c]{} & U_2 \\ \downarrow & & \downarrow \\ U_1 \times U_1 & \xrightarrow{(\mathrm{pr}_1, \mathrm{pr}_2, s_0^0 \circ d_1^1 \circ \mathrm{pr}_1)} & U_1 \times U_1 \times U_1 \end{array}$$

as going around the other way defines a point of U_2. Pulling back the cocycle condition for α on U_2 translates into the condition that the pullbacks of α via the projections to $U_1 \times_{(d_1^1, d_0^1), U_0 \times U_0, (d_1^1, d_0^1)} U_1$ are the same as the pullback of α via $s_0^0 \circ d_1^1 \circ \mathrm{pr}_1$ is the identity map (namely, the pullback of α by s_0^0 is the identity). By Sites, Lemma 04TQ this means that α comes from an isomorphism

$$\alpha' : \mathrm{pr}_1^{-1}\mathcal{G}_0 \to \mathrm{pr}_2^{-1}\mathcal{G}_0$$

of sheaves on $\mathcal{C}/U_0 \times U_0$. Then finally, the morphism $U_2 \to U_0 \times U_0 \times U_0$ is surjective on associated sheaves as is easily seen using the surjectivity of $U_1 \to U_0 \times U_0$ and the description of U_2 given above. Therefore α' satisfies the cocycle condition on $U_0 \times U_0 \times U_0$. The proof is finished by an application of Sites, Lemma 04TS to the covering $\{U_0 \to X\}$. □

Lemma 2.4. *Let $\mathcal{C}$ be a site with equalizers and fibre products. Let K be a hypercovering. The Čech complex of Lemma 0D79 associated to $a^{-1}\mathcal{F}$*

$$a_{0,*}a_0^{-1}\mathcal{F} \to a_{1,*}a_1^{-1}\mathcal{F} \to a_{2,*}a_2^{-1}\mathcal{F} \to \ldots$$

is equal to the complex $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{F})$. Here $s(\mathbf{Z}^{\#}_{F(K)})$ is as in Hypercoverings, Definition 01GB.

Proof. By Lemma 0D86 we have

$$a_{n,*}a_n^{-1}\mathcal{F} = \mathcal{H}om'(F(K_n)^{\#}, \mathcal{F})$$

where $\mathcal{H}om'$ is as in Sites, Section 04TP. The boundary maps in the complex of Lemma 0D79 come from the simplicial structure. Thus the equality of complexes comes from the canonical identifications $\mathcal{H}om'(\mathcal{G}, \mathcal{F}) = \mathcal{H}om(\mathbf{Z}_{\mathcal{G}}, \mathcal{F})$ for $\mathcal{G}$ in $Sh(\mathcal{C})$. □

Proof. First, let $\mathcal{I}$ be an injective abelian sheaf on $\mathcal{C}$. Then the spectral sequence of Lemma 2.1 for the sheaf $a^{-1}\mathcal{I}$ degenerates as $(a^{-1}\mathcal{I})_p = a_p^{-1}\mathcal{I}$ is injective by Lemma 2.2. Thus the complex

$$a_{0,*}a_0^{-1}\mathcal{I} \to a_{1,*}a_1^{-1}\mathcal{I} \to a_{2,*}a_2^{-1}\mathcal{I} \to \ldots$$

computes $Ra_*a^{-1}\mathcal{I}$. By Lemma 2.4 this is equal to the complex $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{I})$. Because K is a hypercovering, we see that $s(\mathbf{Z}^{\#}_{F(K)})$ is exact in degrees > 0 by Hypercoverings, Lemma 01GE applied to the simplicial presheaf $F(K)$. Since $\mathcal{I}$ is injective, the functor $\mathcal{H}om(-, \mathcal{I})$ is exact and we conclude that $\mathcal{H}om(s(\mathbf{Z}^{\#}_{F(K)}), \mathcal{I})$ is exact in positive degrees. We conclude that $R^p a_*a^{-1}\mathcal{I} = 0$ for $p > 0$. On the other hand, we have $\mathcal{I} = a_*a^{-1}\mathcal{I}$ by Lemma 2.3.

Bounded case. Let $E \in D^+(\mathcal{C})$. Choose a bounded below complex $\mathcal{I}^\bullet$ of injectives representing E. By the result of the first paragraph and Leray's acyclicity lemma (Derived Categories, Lemma 015E) $Ra_*a^{-1}\mathcal{I}^\bullet$ is computed by the complex $a_*a^{-1}\mathcal{I}^\bullet = \mathcal{I}^\bullet$ and we conclude the lemma is true in this case.

Unbounded case. We urge the reader to skip this, since the argument is the same as above, except that we use explicit representation by double complexes to get around convergence issues. Let $E \in D(\mathcal{C})$. To show the map $E \to Ra_*a^{-1}E$ is an isomorphism, it suffices to show for every object U of $\mathcal{C}$ that

$$R\Gamma(U, E) = R\Gamma(U, Ra_*a^{-1}E)$$

We will compute both sides and show the map $E \to Ra_*a^{-1}E$ induces an isomorphism. Choose a K-injective complex $\mathcal{I}^\bullet$ representing E. Choose a quasi-isomorphism $a^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ for some K-injective complex $\mathcal{J}^\bullet$ on $(\mathcal{C}/K)_{total}$. We have

$$R\Gamma(U, E) = R\operatorname{Hom}(\mathbf{Z}^\#_U, E)$$

and

$$R\Gamma(U, Ra_*a^{-1}E) = R\operatorname{Hom}(\mathbf{Z}^\#_U, Ra_*a^{-1}E) = R\operatorname{Hom}(a^{-1}\mathbf{Z}^\#_U, a^{-1}E)$$

By Lemma 0D78 we have a quasi-isomorphism

$$\Big(\ldots \to g_{2!}(a_2^{-1}\mathbf{Z}^\#_U) \to g_{1!}(a_1^{-1}\mathbf{Z}^\#_U) \to g_{0!}(a_0^{-1}\mathbf{Z}^\#_U)\Big) \longrightarrow a^{-1}\mathbf{Z}^\#_U$$

Hence $R\operatorname{Hom}(a^{-1}\mathbf{Z}^\#_U, a^{-1}E)$ is equal to

$$R\Gamma((\mathcal{C}/K)_{total}, R\mathcal{H}om(\ldots \to g_{2!}(a_2^{-1}\mathbf{Z}^\#_U) \to g_{1!}(a_1^{-1}\mathbf{Z}^\#_U) \to g_{0!}(a_0^{-1}\mathbf{Z}^\#_U), \mathcal{J}^\bullet))$$

By the construction in Cohomology on Sites, Section 08J7 and since $\mathcal{J}^\bullet$ is K-injective, we see that this is represented by the complex of abelian groups with terms

$$\prod\nolimits_{p+q=n} \operatorname{Hom}(g_{p!}(a_p^{-1}\mathbf{Z}^\#_U), \mathcal{J}^q) = \prod\nolimits_{p+q=n} \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, g_p^{-1}\mathcal{J}^q)$$

See Cohomology on Sites, Lemmas 0A94 and 08JA for more information. Thus we find that $R\Gamma(U, Ra_*a^{-1}E)$ is computed by the product total complex $\operatorname{Tot}_\pi(B^{\bullet,\bullet})$ with $B^{p,q} = \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, g_p^{-1}\mathcal{J}^q)$. For the other side we argue similarly. First we note that

$$s(\mathbf{Z}^\#_{F(K)}) \longrightarrow \mathbf{Z}$$

is a quasi-isomorphism of complexes on $\mathcal{C}$ by Hypercoverings, Lemma 01GE. Since $\mathbf{Z}^\#_U$ is a flat sheaf of $\mathbf{Z}$-modules we see that

$$s(\mathbf{Z}^\#_{F(K)}) \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U \longrightarrow \mathbf{Z}^\#_U$$

is a quasi-isomorphism. Therefore $R\operatorname{Hom}(\mathbf{Z}^\#_U, E)$ is equal to

$$R\Gamma(\mathcal{C}, R\mathcal{H}om(s(\mathbf{Z}^\#_{F(K)}) \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U, \mathcal{I}^\bullet))$$

By the construction of $R\mathcal{H}om$ and since $\mathcal{I}^\bullet$ is K-injective, this is represented by the complex of abelian groups with terms

$$\prod\nolimits_{p+q=n} \operatorname{Hom}(\mathbf{Z}^\#_{K_p} \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U, \mathcal{I}^q) = \prod\nolimits_{p+q=n} \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}\mathcal{I}^q)$$

The equality of terms follows from the fact that $\mathbf{Z}^\#_{K_p} \otimes_{\mathbf{Z}} \mathbf{Z}^\#_U = a_{p!}a_p^{-1}\mathbf{Z}^\#_U$ by Modules on Sites, Remark 0EYY. Thus we find that $R\Gamma(U, E)$ is computed by the product total complex $\operatorname{Tot}_\pi(A^{\bullet,\bullet})$ with $A^{p,q} = \operatorname{Hom}(a_p^{-1}\mathbf{Z}^\#_U, a_p^{-1}\mathcal{I}^q)$.

Since $\mathcal{I}^\bullet$ is K-injective we see that $a_p^{-1}\mathcal{I}^\bullet$ is K-injective, see Lemma 2.2. Since $\mathcal{J}^\bullet$ is K-injective we see that $g_p^{-1}\mathcal{J}^\bullet$ is K-injective, see Lemma 09WH. Both represent the object $a_p^{-1}E$. Hence for every $p \geq 0$ the map of complexes

$$A^{p,\bullet} = \mathrm{Hom}(a_p^{-1}\mathbf{Z}_U^\#, a_p^{-1}\mathcal{I}^\bullet) \longrightarrow \mathrm{Hom}(a_p^{-1}\mathbf{Z}_U^\#, g_p^{-1}\mathcal{J}^\bullet) = B^{p,\bullet}$$

induced by g_p^{-1} applied to the given map $a^{-1}\mathcal{I}^\bullet \to \mathcal{J}^\bullet$ is a quasi-isomorphisms as these complexes both compute

$$R\,\mathrm{Hom}(a_p^{-1}\mathbf{Z}_U^\#, a_p^{-1}E)$$

By More on Algebra, Lemma 0EYX we conclude that the right vertical arrow in the commutative diagram

$$\begin{array}{ccc} R\Gamma(U, E) & \longrightarrow & \mathrm{Tot}_\pi(A^{\bullet,\bullet}) \\ \downarrow & & \downarrow \\ R\Gamma(U, Ra_*a^{-1}E) & \longrightarrow & \mathrm{Tot}_\pi(B^{\bullet,\bullet}) \end{array}$$

is a quasi-isomorphism. Since we saw above that the horizontal arrows are quasi-isomorphisms, so is the left vertical arrow. □

Lemma 2.5. *Let $\mathcal{C}$ be a site. Let K be a simplicial presheaf. Let $\mathcal{G}$ be a presheaf. Let $K \to \mathcal{G}$ be an augmentation of K towards $\mathcal{G}$. Assume that*

(1) *The morphism of presheaves $K_0 \to \mathcal{G}$ becomes a surjection after sheafification.*
(2) *The morphism*
$$(d_0^1, d_1^1) : K_1 \longrightarrow K_0 \times_{\mathcal{G}} K_0$$
becomes a surjection after sheafification.
(3) *For every $n \geq 1$ the morphism*
$$K_{n+1} \longrightarrow (cosk_n sk_n K)_{n+1}$$
turns into a surjection after sheafification.

Then $H_i(K) = 0$ for $i > 0$ and $H_0(K) = \mathbf{Z}_{\mathcal{G}}^\#$.

Proof. Denote $K^n = \mathrm{cosk}_n \mathrm{sk}_n K$ for $n \geq 1$. Define K^0 as the simplicial object with terms $(K^0)_n$ equal to the $(n+1)$-fold fibred product $K_0 \times_{\mathcal{G}} \ldots \times_{\mathcal{G}} K_0$, see Simplicial, Example 016E. We have morphisms

$$K \longrightarrow \ldots \to K^n \to K^{n-1} \to \ldots \to K^1 \to K^0.$$

The morphisms $K \to K^i$, $K^j \to K^i$ for $j \geq i \geq 1$ come from the universal properties of the cosk_n functors. The morphism $K^1 \to K^0$ is the canonical morphism from Simplicial, Remark 018J. We also recall that $K^0 \to \mathrm{cosk}_1 \mathrm{sk}_1 K^0$ is an isomorphism, see Simplicial, Lemma 018I.

By Lemma 01GC we see that $H_i(K^0) = 0$ for $i > 0$ and $H_0(K^0) = \mathbf{Z}_{\mathcal{G}}^\#$.

Pick $n \geq 1$. Consider the morphism $K^n \to K^{n-1}$. It is an isomorphism on terms of degree $< n$. Note that $K^n \to \mathrm{cosk}_n \mathrm{sk}_n K^n$ and $K^{n-1} \to \mathrm{cosk}_n \mathrm{sk}_n K^{n-1}$ are isomorphisms. Note that $(K^n)_n = K_n$ and that $(K^{n-1})_n = (\mathrm{cosk}_{n-1} \mathrm{sk}_{n-1} K)_n$. Hence by assumption, we have that $(K^n)_n \to (K^{n-1})_n$ is a morphism of presheaves which becomes surjective after sheafification. By Lemma 01GD we conclude that $H_i(K^n) = H_i(K^{n-1})$. Combined with the above this proves the lemma. □